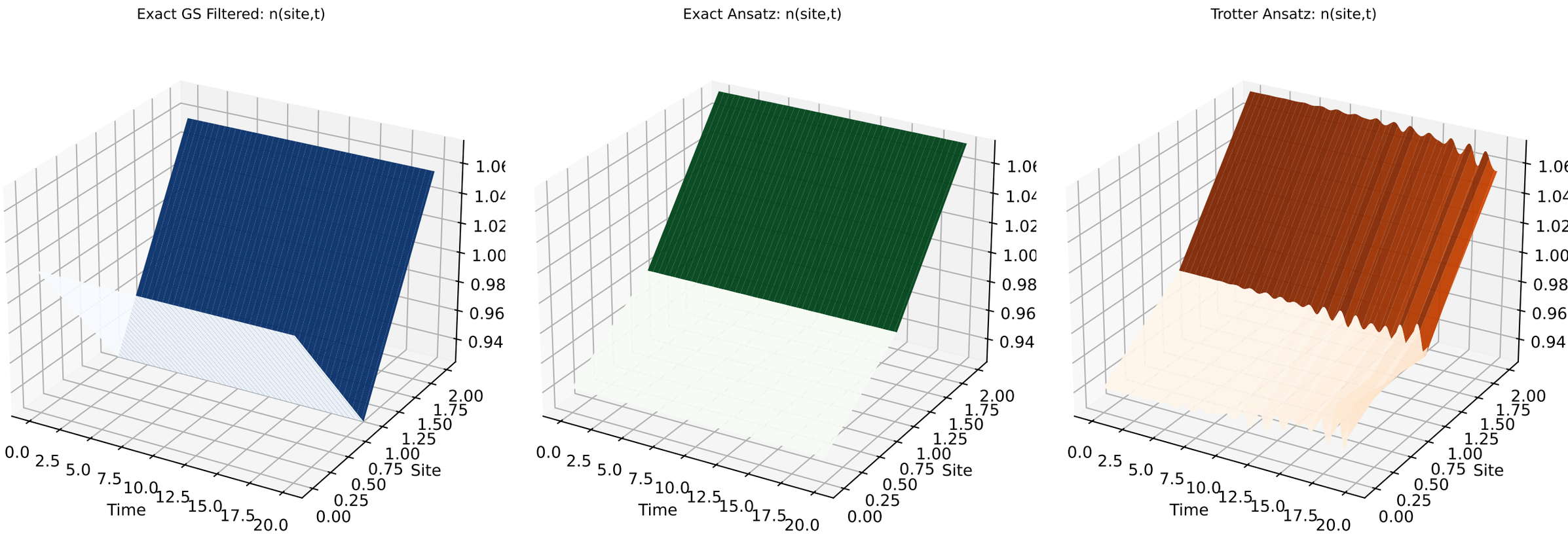
Executed Command

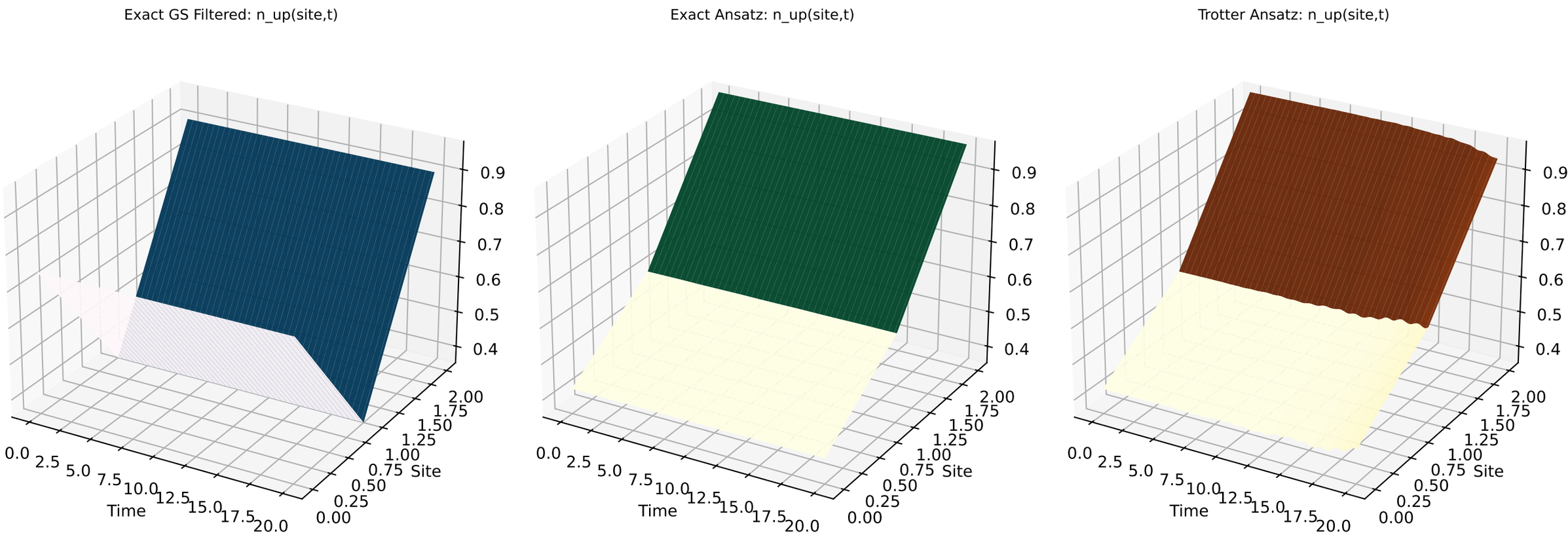
Reference: pipelines/PIPELINE_RUN_GUIDE.md
Script: pipelines/hardcoded_hubbard_pipeline.py

```
/opt/anaconda3/bin/python pipelines/hardcoded_hubbard_pipeline.py --L 3 --t 1.0 --u 4.0 --dv 0.0 --boundary
  periodic --ordering blocked --t-final 20.0 --num-times 401 --suzuki-order 2 --trotter-steps 128 --term-order
  sorted --vqe-reps 2 --vqe-restarts 3 --vqe-maxiter 600 --qpe-eval-qubits 5 --qpe-shots 256 --qpe-seed 11
  --initial-state-source vqe --output-json artifacts/json/H_L3_static_t1.0_U4.0_S128.json --output-pdf
  artifacts/pdf/H_L3_static_t1.0_U4.0_S128.pdf
```

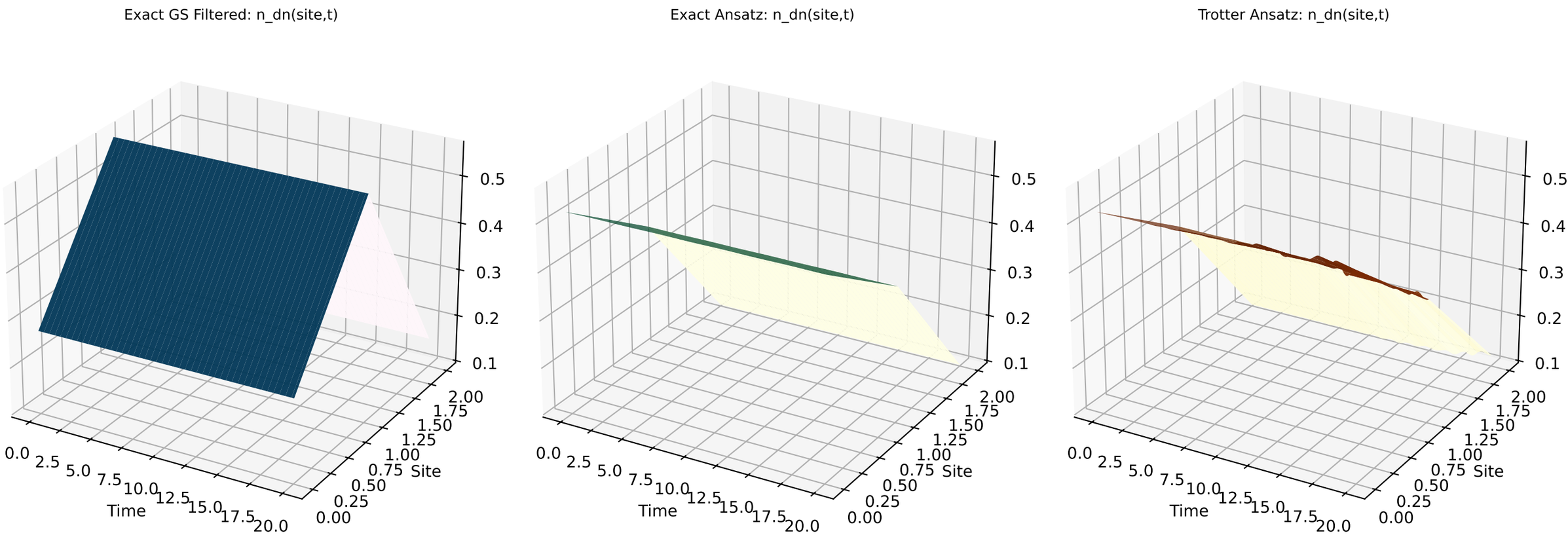
L=3 3D Densities (Total n)



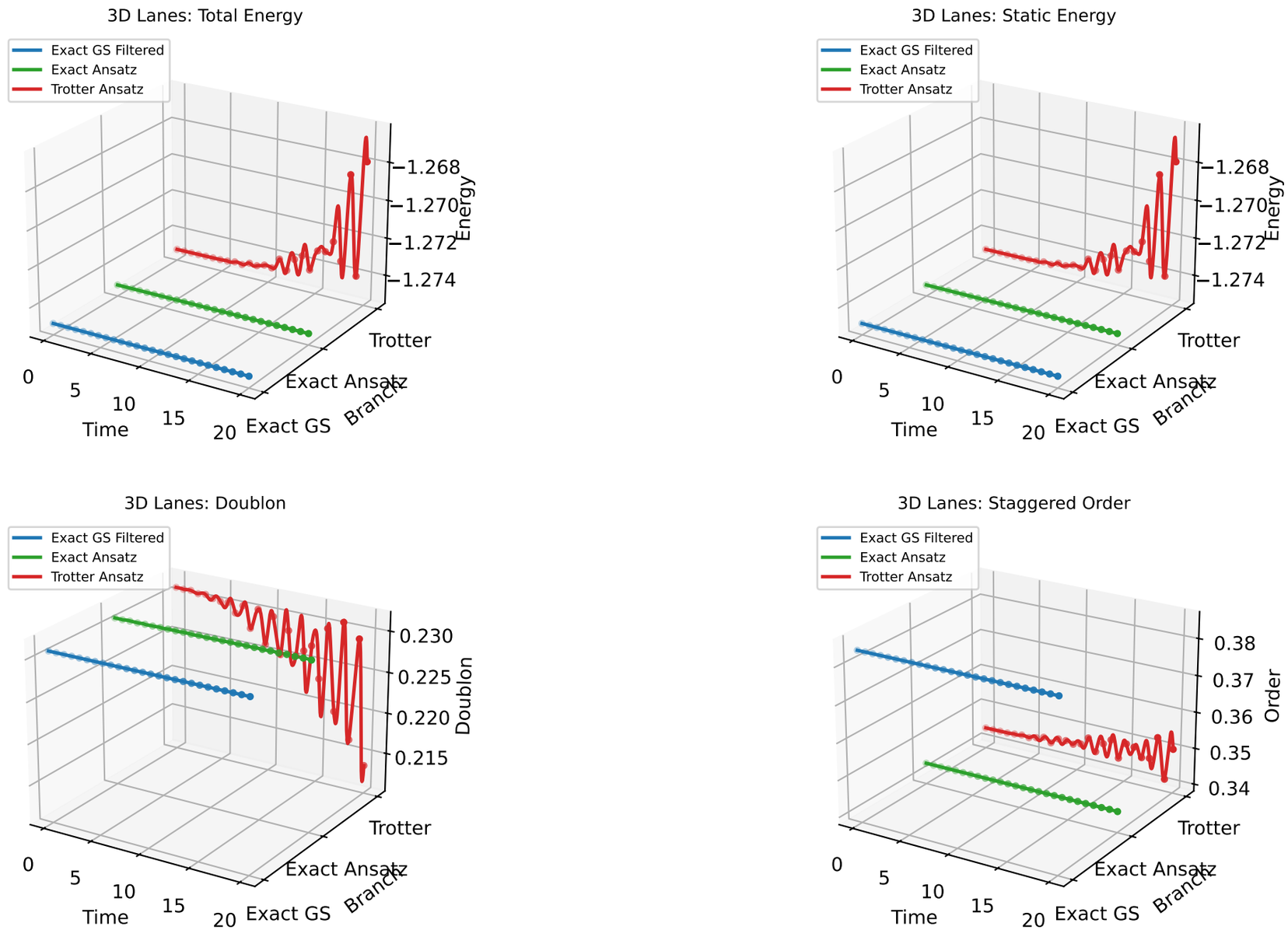
L=3 3D Densities (Spin-Up)

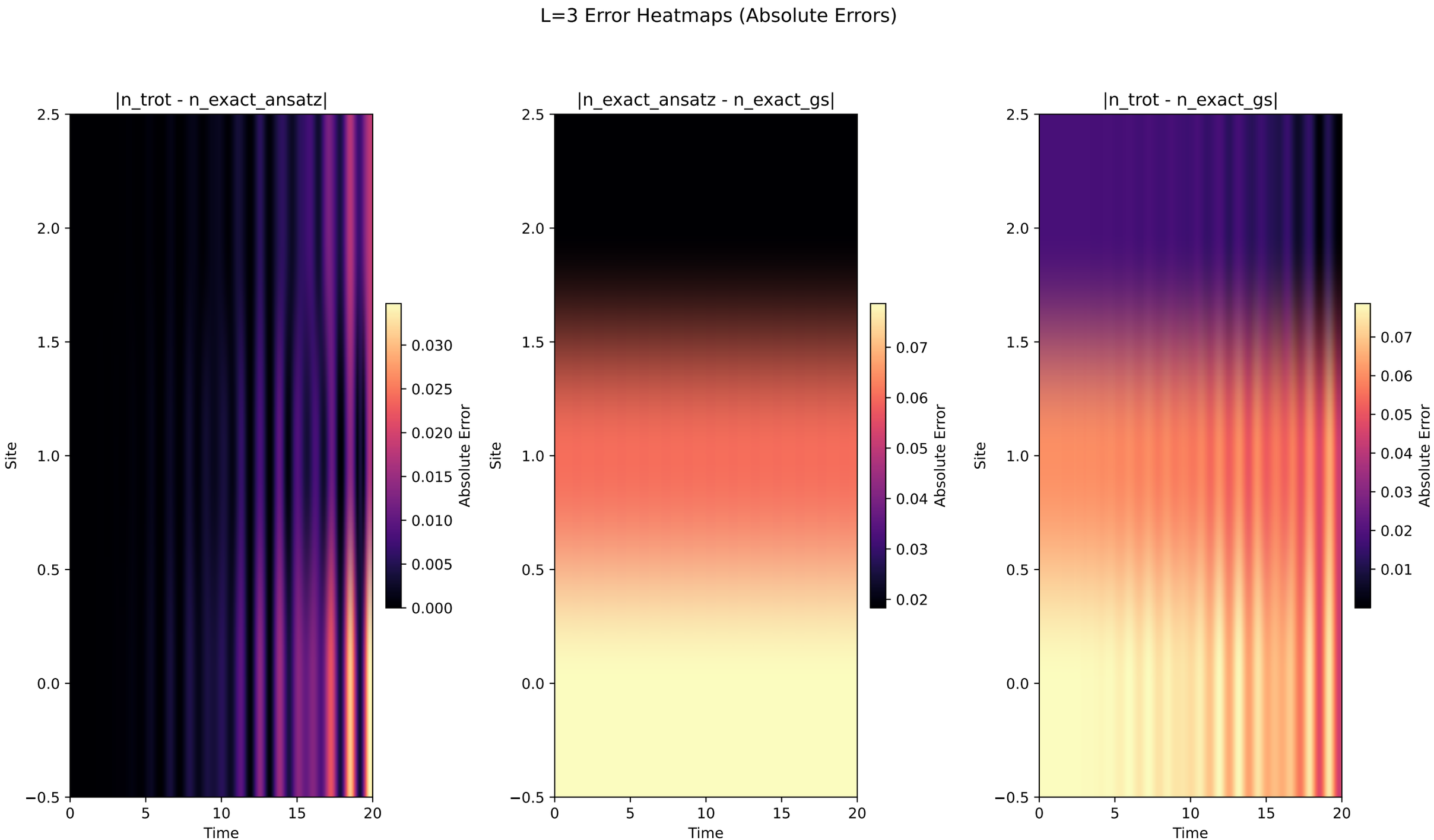


L=3 3D Densities (Spin-Down)

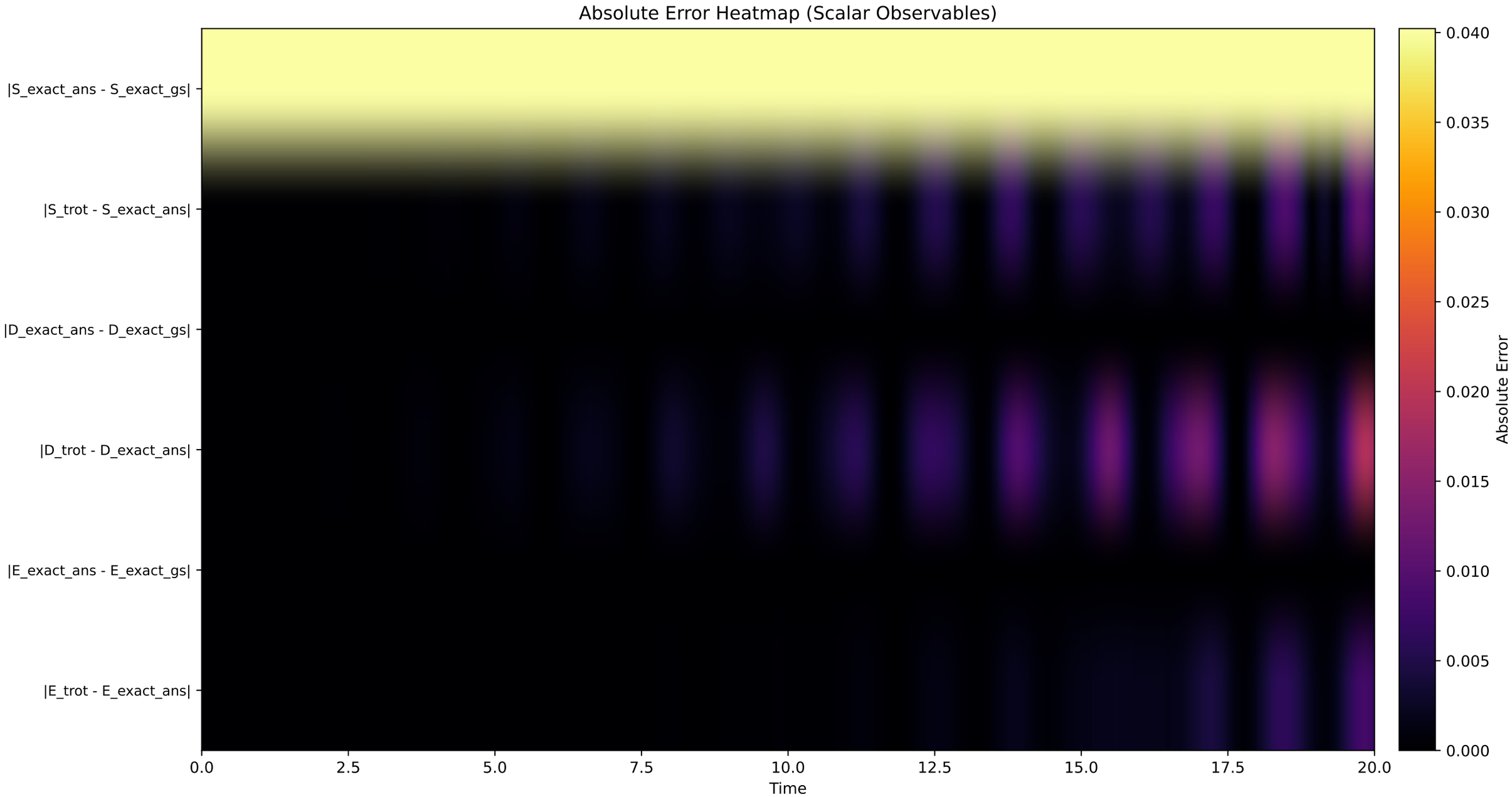


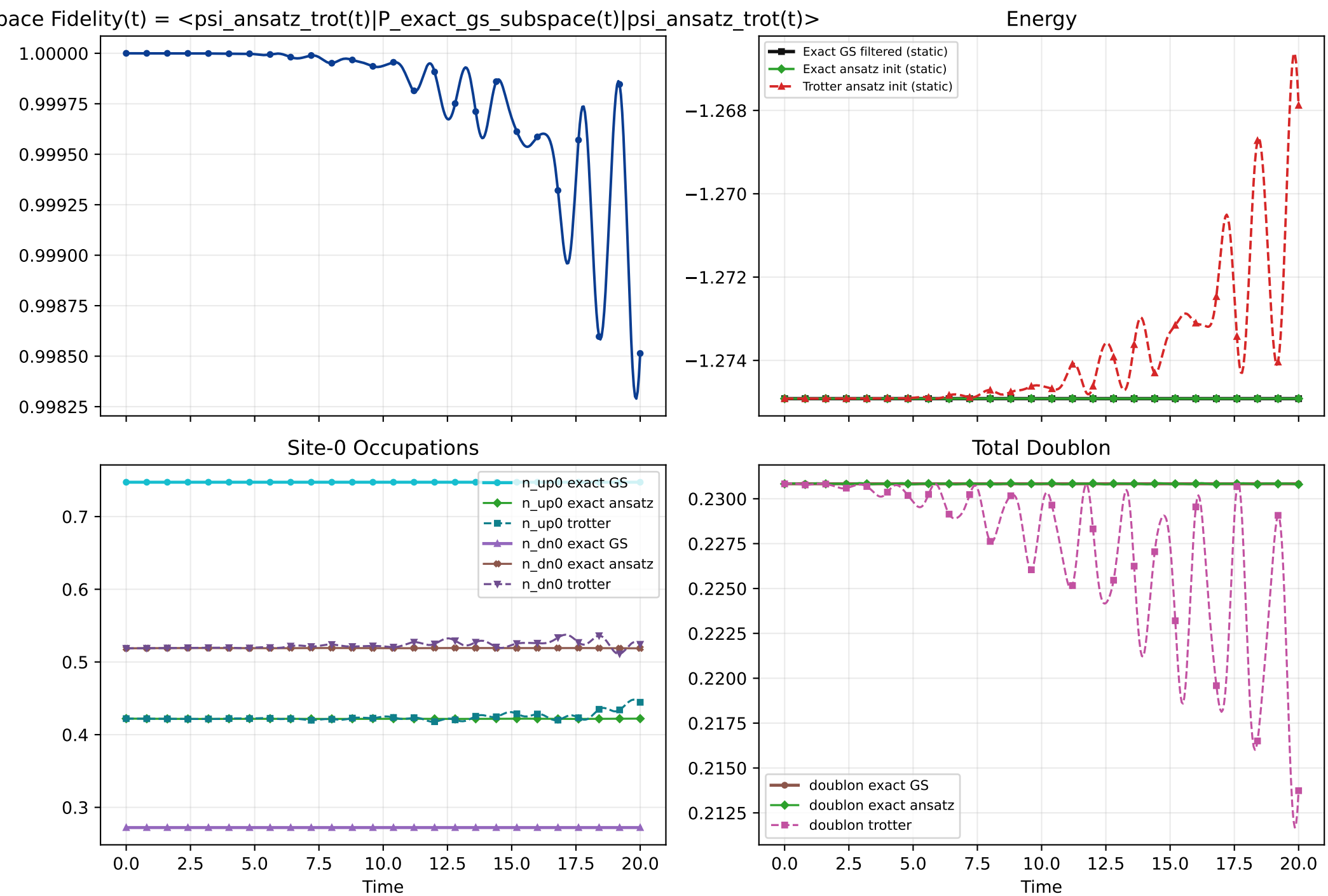
L=3 3D Scalar Observables (Three Evolutions)

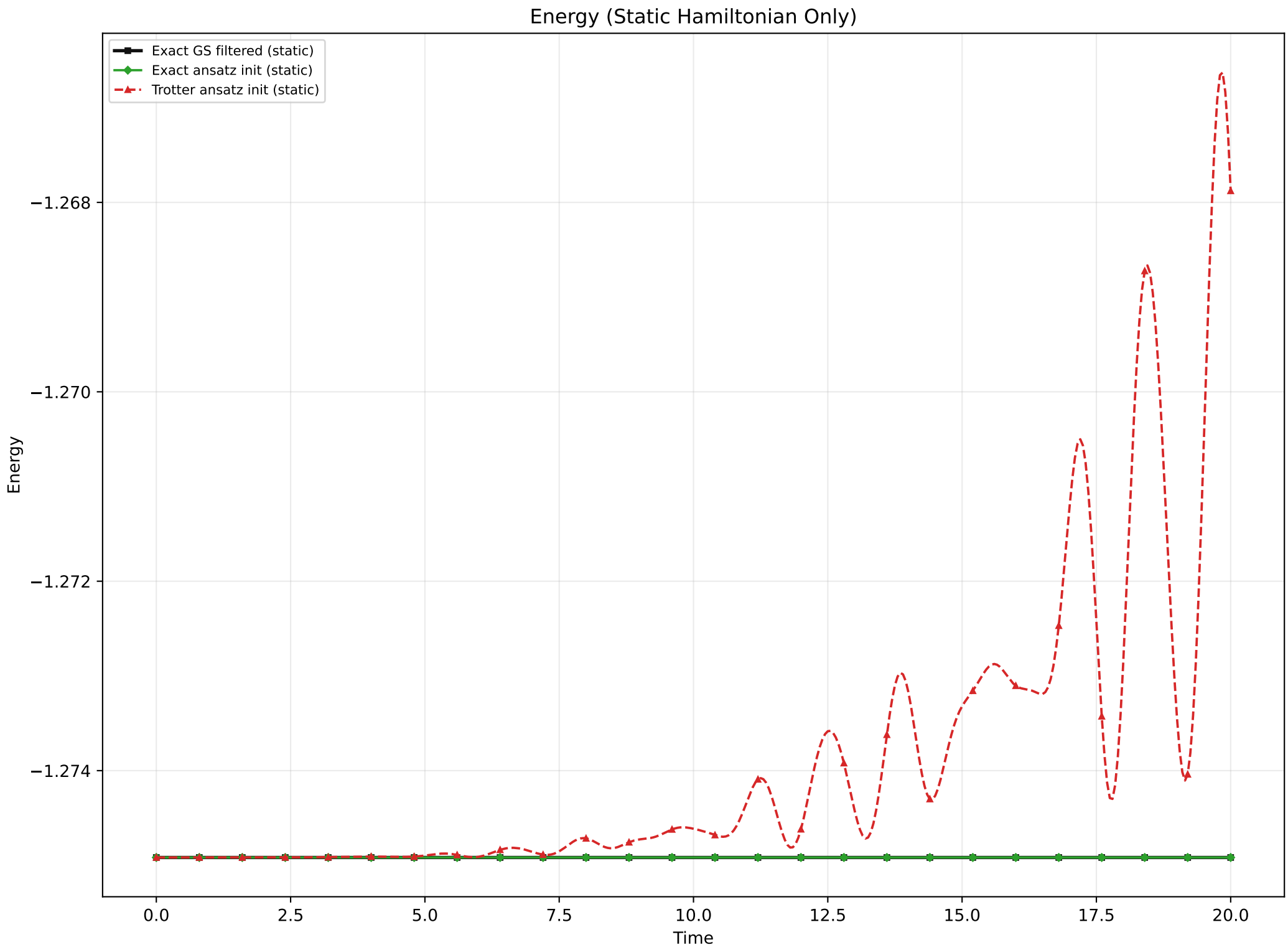


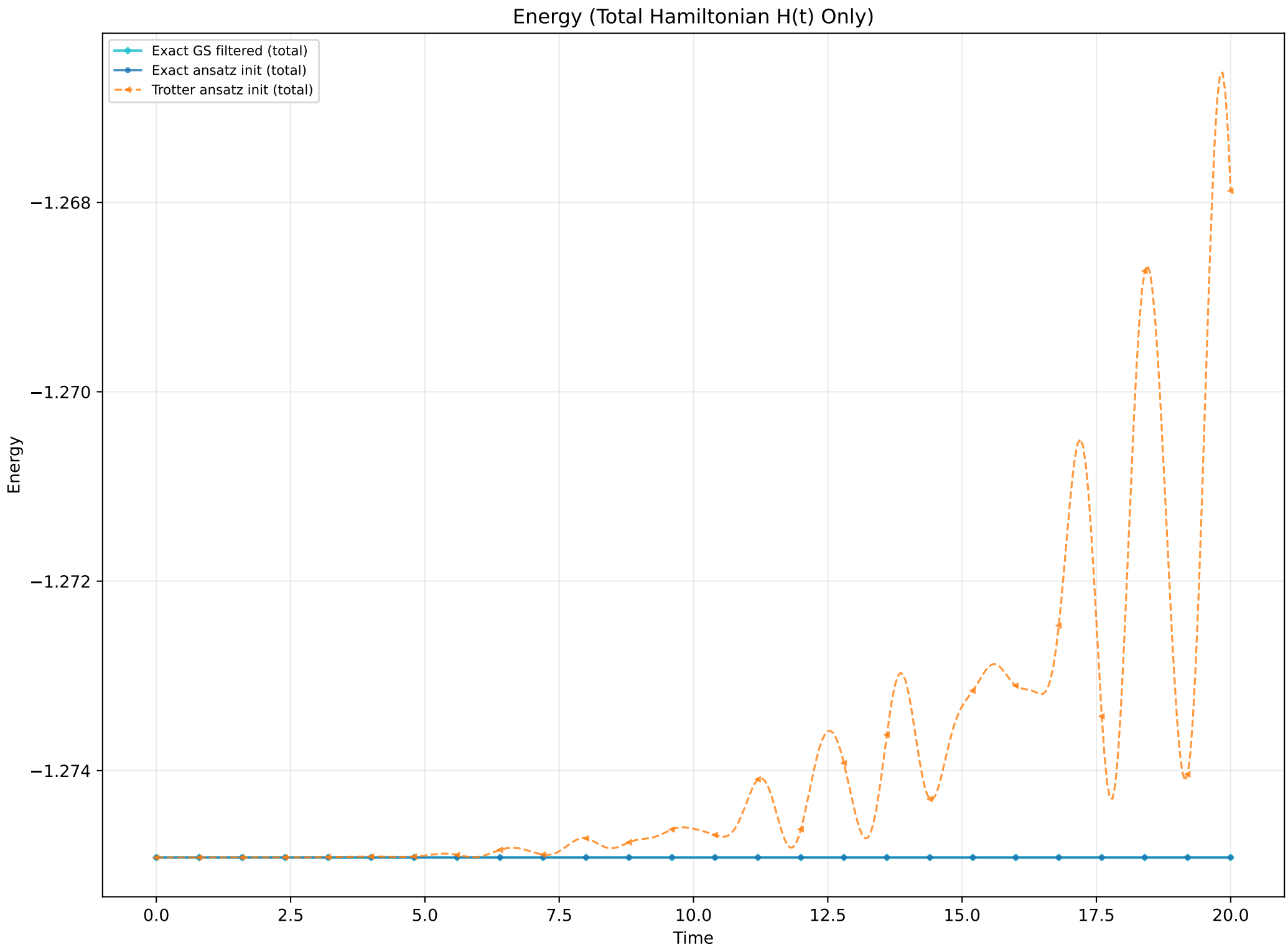

L=3 Error Heatmaps (Absolute Errors)
|n_trot - n_exact_ansatz|
|n_exact_ansatz - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
5
10
15
20
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0.07
0.06
0.05
0.04
0.03
0.02
0.01

L=3 Scalar Error Heatmap




Hardcoded Hubbard Pipeline: L=3
pace Fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>
1.00000
0.99975
0.99950
0.99925
0.99900
0.99875
0.99850
0.99825
Energy
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)
−1.268
−1.270
−1.272
−1.274
Site-0 Occupations
n_up0 exact GS
n_up0 exact ansatz
n_up0 trotter
n_dn0 exact GS
n_dn0 exact ansatz
n_dn0 trotter
0.7
0.6
0.5
0.4
0.3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time
Total Doublon
0.2300
0.2275
0.2250
0.2225
0.2200
0.2175
0.2150
0.2125
doublon exact GS
doublon exact ansatz
doublon trotter
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time

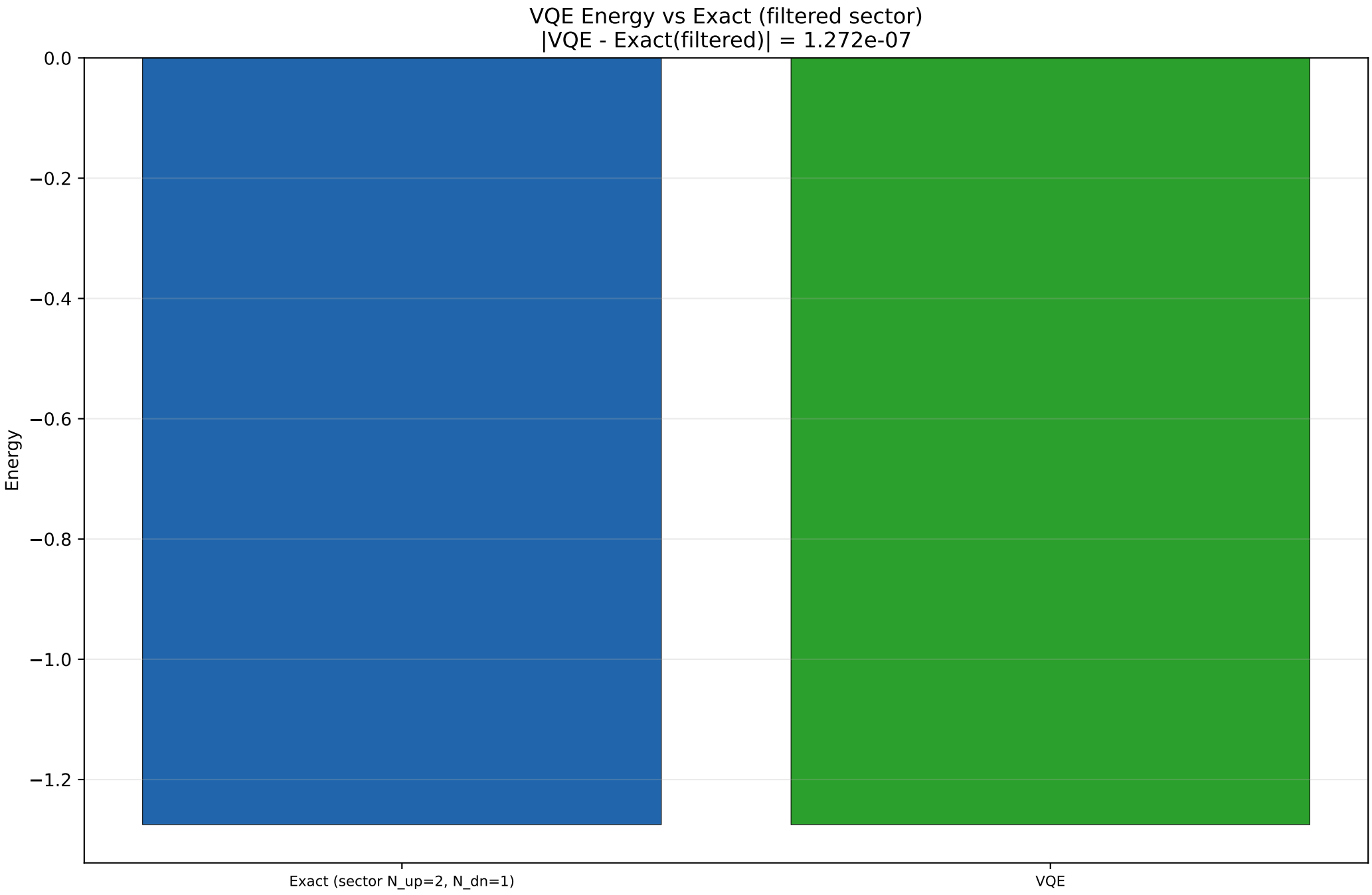

Energy (Static Hamiltonian Only)
Exact GS filtered (static)
Exact ansatz init (static)
Trotter ansatz init (static)
−1.268
−1.270
−1.272
−1.274
Energy
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time


Energy (Total Hamiltonian H(t) Only)
Exact GS filtered (total)
Exact ansatz init (total)
Trotter ansatz init (total)
−1.268
−1.270
−1.272
−1.274
Energy
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time

Appendix — L=3

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

settings: {"L": 3, "t": 1.0, "u": 4.0, "dv": 0.0, "boundary": "periodic", "ordering": "blocked", "t_final": 20.0, "num_times":
exact_trajectory_label: Exact_Hardcode
exact_trajectory_method: python_matrix_eigendecomposition
fidelity_definition: fidelity(t) = <psi_ansatz_trot(t)|P_exact_gs_subspace(t)|psi_ansatz_trot(t)>, where P_exact_gs_subspace(t
subspace_fidelity_at_t0: 0.9999999180073835
energy_t0_exact_gs: -1.2749172176353754
energy_t0_exact_ansatz: -1.2749170904348224
energy_t0_trotter: -1.2749170904348222
ground_state_exact_energy (full Hilbert): -3.123105625618
ground_state_exact_energy_filtered: -1.2749172176353751
filtered_sector: {'n_up': 2, 'n_dn': 1}
vqe_energy: -1.27491709043483
qpe_energy_estimate: -1.125
initial_state_source: vqe
hamiltonian_terms: 22
reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```